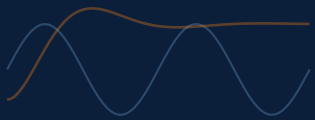
DAY 2 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



System Dynamics and Control

Linearisation ▪ Transfer functions & state space ▪ Identification ▪ PID ▪ Stability

Mechatronics Training Programme

Theoretical Course Series ▪ Day 2 of 5

Agenda and learning objectives

Code	Session	Time
2.1	Nonlinear effects and linearisation	0.5 h
2.2	Transfer functions and state space	1.0 h
2.3	System identification and frequency response	1.0 h
2.4	PID and basic control structures	1.5 h
2.5	Stability and performance analysis	1.0 h
	Net teaching time (+ short breaks)	**5.0 h**

Refresher boxes today: complex numbers, Laplace transform, matrices, sinusoidal response, decibels. **"Going further"** items (small grey text) are optional.

By the end of today you can . . .

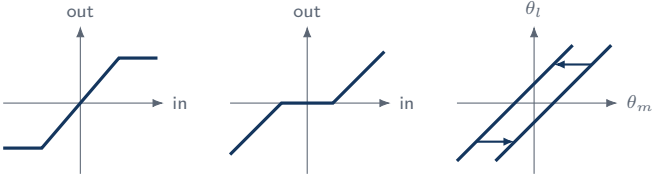
- **explain** why a linear model is only valid near an operating point
- **turn** a simple ODE into a transfer function and **find** its poles
- **read** a Bode plot (dB, phase) and **fit** a simple model to a step test
- **explain** P, I and D with the cruise-control example and **tune** a PI with a simple rule
- **judge** stability from pole positions and from gain/phase margins

Session 1

2.1 Nonlinear Effects and Linearisation

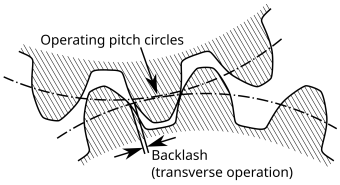
What “nonlinear” means: limits, gaps and play

Linear = doubling the input doubles the output. Real machines break this rule:



Saturation **Dead-zone** **Backlash**

Effect	Everyday picture	In a servo axis
Saturation	gas pedal already on the floor	voltage / current limit
Dead-zone	loose steering wheel	IGBT dead-time
Backlash	gap between gear teeth	gears, belts, couplings
Magn. saturation	a sponge that is already full	motor iron at high current

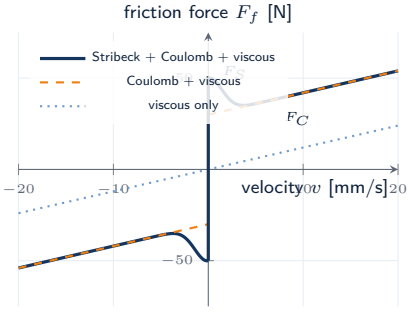


GearHeads, Slashme, CC BY-SA 3.0, via Wikimedia Commons

Magnetic saturation: at high current the motor's **inductance** drops (e.g. 10 mH → 3 mH), so the current loop becomes “livelier”.

Going further: describing functions predict limit cycles [Slotine & Li, 1991].

Friction: hard to start, easier to keep moving



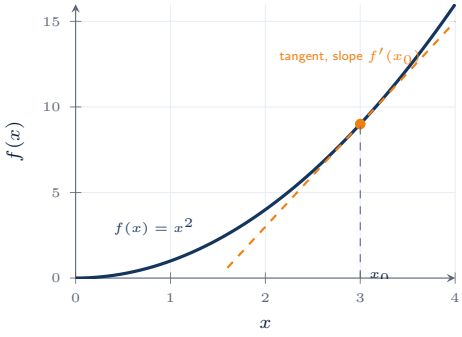
Think of pushing a heavy wardrobe: it needs a big push to start, then less to keep it sliding.

$$F_f(v) = \underbrace{F_C \operatorname{sgn}(v)}_{\text{Coulomb}} + \underbrace{B\,v}_{\text{viscous}}$$

- F_S breakaway force [N], F_C Coulomb level [N], B viscous coefficient [N s/m].
- At $v = 0$ the force jumps: **no slope** exists, so it cannot be linearised.
- Result in practice: **stick–slip** and "hunting" around the set-point.
- Fix: add the expected friction force as feedforward.

Going further: Stribeck curve $F_C + (F_S - F_C)e^{-(v/v_S)^2}$, LuGre model. [Armstrong-Hélouvry et al., Automatica 1994]

Linearisation: zoom in until the curve looks straight



Close to an **operating point** x_0 every smooth curve looks like its tangent line.

$$f(x) \approx f(x_0) + f'(x_0)\,(x - x_0)$$

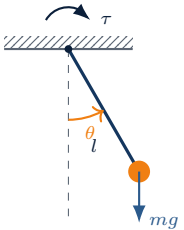
Example $f = x^2$, $x_0 = 3$:

1. slope $f'(3) = 2 \cdot 3 = 6$
2. at $x = 3.1$: linear $9 + 6 \cdot 0.1 = 9.6$, exact 9.61 (0.1 % error)
3. at $x = 4$: linear 15, exact 16 (6 % error) $\Rightarrow$ only **local**

$\delta x = x - x_0$ is the small **deviation** from the operating point.

Going further: many variables: $\delta\dot{\boldsymbol{x}} = A\,\delta\boldsymbol{x} + B\,\delta\boldsymbol{u}$ with Jacobians $A = \partial\boldsymbol{f}/\partial\boldsymbol{x}$, $B = \partial\boldsymbol{f}/\partial\boldsymbol{u}$; eigenvalues of A give local stability (Lyapunov's indirect method).

Example 1: pendulum / unbalanced arm on a servo motor



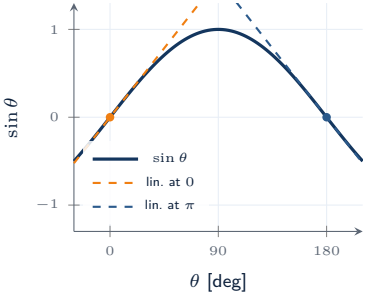
$J\ddot{\theta} = \tau - mgl\sin\theta - b\dot{\theta}$ (J [kg m^2], τ [Nm], b [Nm s/rad])

1. Only $\sin\theta$ is nonlinear; slope of $\sin$ at θ_0 is $\cos\theta_0$.
2. Linear model:

$$J\,\delta\ddot{\theta} + b\,\delta\dot{\theta} + (mgl\cos\theta_0)\,\delta\theta = \delta\tau$$

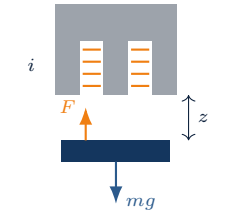
3. **Hanging** ($\cos 0 = +1$): gravity acts like a spring $\Rightarrow$ **stable** swing.
4. **Inverted** ($\cos\pi = -1$): a "negative spring" $\Rightarrow$ **falls over**.

Numbers: $m = 2\,\text{kg}$, $l = 0.3\,\text{m}$, $J = ml^2 \Rightarrow$ $\omega_n = \sqrt{g/l} = 5.7\,\text{rad/s}$ (0.9 Hz).



Error of $\sin\theta \approx \theta$: 1 % at 14°, 5 % at 31°.

Example 2: electromagnet (magnetic levitation / actuator)



Svjo, CC BY-SA 3.0, via Wikimedia Commons

Magnet force $F = k\,i^2/z^2$ (current i [A], air gap z [m]). Like balancing a broom: the closer the iron, the **stronger** the pull.

$$\delta F = K_i\,\delta i - K_z\,\delta z \quad \Rightarrow \quad \frac{\delta Z(s)}{\delta I(s)} = \frac{-K_i}{ms^2 - K_z}$$

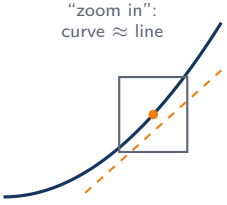
Worked example: $m = 0.1\,\text{kg}$, gap $z_0 = 5\,\text{mm}$, current $i_0 = 1\,\text{A}$.

1. Weight: $mg = 0.98\,\text{N}$ must equal the magnet force at (z_0, i_0).
2. Slopes: $K_z = 2mg/z_0 = 392\,\text{N/m}$ (**negative** stiffness), $K_i = 2mg/i_0 = 1.96\,\text{N/A}$.
3. Poles: $s = \pm\sqrt{K_z/m} = \pm 62.6\,\text{s}^{-1} \Rightarrow$ one pole in the right half-plane: **unstable**.
4. $K_z \propto 1/z_0$: at another gap the model changes $\Rightarrow$ valid only near (z_0, i_0).

Session 2.1 – Check your understanding

Check your understanding

1. Why can we *not* linearise Coulomb friction at $v = 0$?
2. The hanging and the inverted pendulum obey the *same* equation. Why is one stable and the other not?
3. The maglev model was derived at $z_0 = 5\,\text{mm}$. Can you use it at $z = 10\,\text{mm}$? What changes?



Discuss in pairs for 2 minutes.

Session 2.1 – Summary and key terms

Key take-aways

- Real axes have saturation, dead-zone, backlash, friction and magnetic saturation.
- Smooth curves are **linearised** with the tangent at an operating point.
- The linear model is **local**: gains and even stability can change with the operating point.
- Jumps (friction at $v = 0$, backlash) need compensation or simulation instead.

Key terms

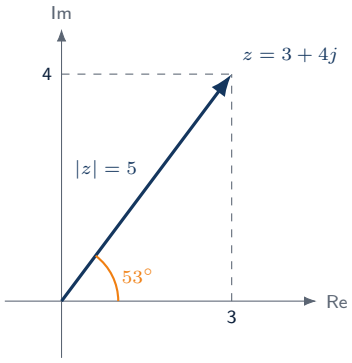
Operating point: steady condition we linearise around.
Deviation δx: small change from that point.
Saturation: output stops growing at a limit.
Backlash: play/gap in a transmission.
Stick–slip: jerky motion caused by friction.

Next: with a linear model in hand, we describe it compactly by a transfer function or a state-space model.

Session 2

2.2 Transfer Functions and State Space

Refresher: complex numbers and the complex plane



Refresher: complex numbers

$z = a + jb$ (engineers write $j = \sqrt{-1}$). Polar form $z = |z|\, e^{j\varphi}$: $|z| = \sqrt{a^2 + b^2}$, $\varphi = \angle z = \operatorname{atan2}(b, a)$.

$$e^{j\varphi} = \cos\varphi + j\sin\varphi \qquad \text{(Euler)}$$

- **Multiply**: magnitudes multiply, angles add. **Divide**: magnitudes divide, angles subtract.
- Example: $\dfrac{1}{1+j}$: $|1 + j| = 1.414$, $\angle = 45° \Rightarrow 0.707\angle -45°$.
- For $s = \sigma + j\omega$: $e^{st} = e^{\sigma t}(\cos\omega t + j\sin\omega t) \Rightarrow \sigma$ sets growth/decay, ω sets oscillation.

Refresher: the Laplace transform as a tool

Refresher: Laplace transform

A "translation" $f(t) \to F(s)$ that turns **differential equations into algebra** – like logarithms turn multiplication into addition. We only use the table.

Time domain $f(t)$	Laplace $F(s)$
unit impulse $\delta(t)$	1
unit step $1(t)$	$1/s$
ramp t	$1/s^2$
e^{-at}	$1/(s+a)$
$\sin \omega t$	$\omega/(s^2+\omega^2)$
derivative $\dot{f}$	$sF(s) - f(0)$
integral $\int_0^t f\, d\tau$	$F(s)/s$
delay $f(t-T)$	$e^{-sT}F(s)$

Worked example: RC circuit, $RC = 1\,\text{ms}$, input u, capacitor voltage v (start at 0 V):

1. ODE: $RC\,\dot{v} + v = u$
2. Transform ($\dot{v} \to sV$): $RC\,s\,V + V = U$
3. Solve for the ratio:

$$\frac{V(s)}{U(s)} = \frac{1}{RC\,s+1} = \frac{1}{0.001\,s+1}$$

4. 1 V step: $V = \frac{1}{s(0.001s+1)} \Rightarrow$ table: $v(t) = 1 - e^{-t/1\,\text{ms}}$ V – the familiar RC charging curve.

Transfer function, poles and zeros

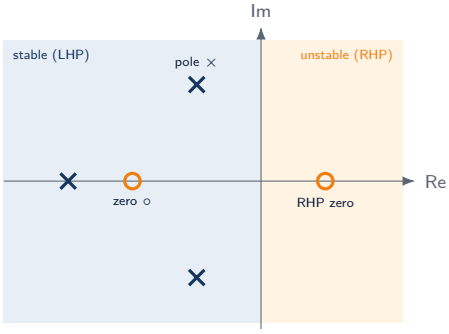
The **transfer function** is the "recipe" of a system: output over input, with zero initial conditions.

$$G(s) = \frac{Y(s)}{U(s)} = \frac{\text{numerator}(s)}{\text{denominator}(s)}$$

- **Poles**: roots of the denominator $\Rightarrow$ decide **stability and speed**.
- **Zeros**: roots of the numerator $\Rightarrow$ shape the response.
- **Order** = number of poles = number of energy stores (inductors, masses, ...).
- **DC gain** $G(0)$: output/input after everything has settled.

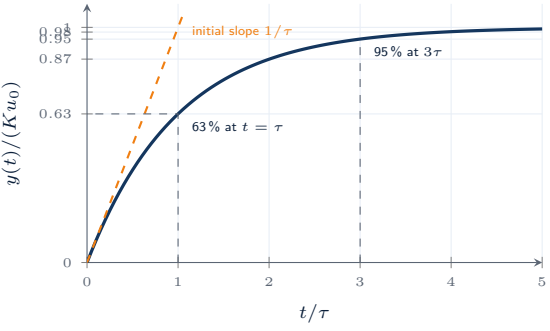
Example: $G(s) = \dfrac{s+2}{(s+1)(s+3)}$: zero -2; poles $-1, -3$; $G(0) = 2/3$.

Going further: RHP zeros (undershoot), delay e^{-sT} is not a polynomial ratio (Padé approximation).



Left half-plane (LHP) = stable side.

First-order system: one energy store

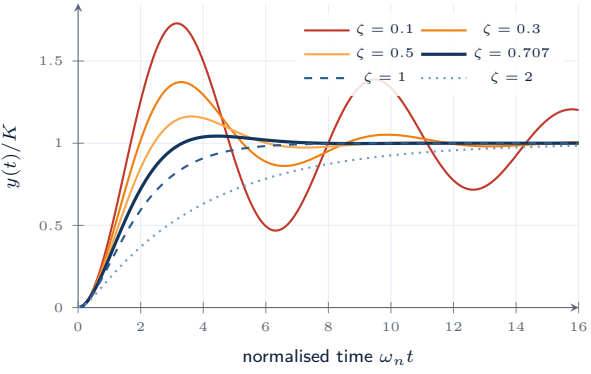


Like a cup of coffee cooling or an RC circuit charging.

$$G(s) = \frac{K}{\tau s + 1}, \quad y(t) = K u_0 \left(1 - e^{-t/\tau}\right)$$

- K gain, τ **time constant** [s]; one pole at $s = -1/\tau$.
- 63 % after τ, $\approx$ 98 % after 4τ; never overshoots.
- Motor winding: $I/V = 1/(L_a s + R)$, $\tau_e = L_a/R$ (e.g. 2 mH/1 Ω = 2 ms).
- Flywheel with friction: $\Omega/T = 1/(Js + B)$, $\tau = J/B$.

Second-order system: the car suspension



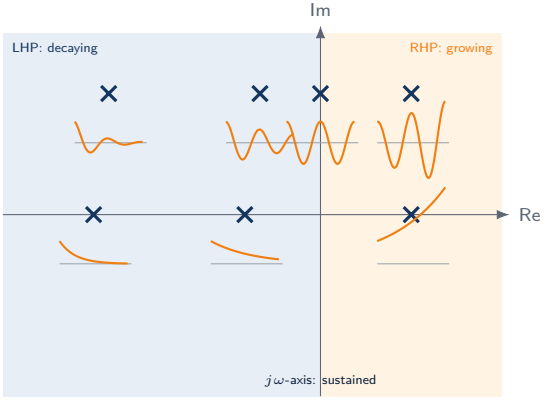
Mass m, spring k, damper c:
$m\ddot{x} + c\dot{x} + kx = F$.

$$G(s) = \frac{K\,\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- **Natural frequency** $\omega_n = \sqrt{k/m}$ [rad/s].
- **Damping ratio** $\zeta = c/(2\sqrt{km})$ [–].
- Small ζ: bouncy (worn shock absorbers); $\zeta \geq 1$: no overshoot but slow.
- $\zeta \approx 0.7$: fast with $\approx 5\,\%$ overshoot.

Quarter car: $m = 250\,\text{kg}$, $k = 16\,\text{kN/m}$, $c = 2000\,\text{N s/m} \Rightarrow \omega_n = 8\,\text{rad/s}$, $\zeta = 0.5$.

Where the poles are tells you how the system moves



A pole $p = -\sigma \pm j\omega_d$ gives a motion $e^{-\sigma t}\cos(\omega_d t)$:

- **Real part** $-\sigma$: how fast it dies out; settles in $\approx 4/\sigma$.
- **Imaginary part** ω_d: how fast it wiggles [rad/s].
- Right half-plane: grows $\Rightarrow$ **unstable**.
- Slowest poles (nearest the $j\omega$-axis) **dominate**.

Example: poles -2 and -20: the pole at -2 dominates; settling $\approx 4/2 = 2$ s.

For the 2nd-order system:
$p = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$, $|p| = \omega_n$.

Refresher: matrices for state space

Refresher: matrix times vector

Each row of A times the vector $\boldsymbol{x}$ gives one entry:

$$\begin{bmatrix} 0 & 1 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 2 \\ -4 \cdot 1 - 2 \cdot 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$$

Why? Two coupled ODEs

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = -4x_1 - 2x_2$$

fit in one line: $\dot{\boldsymbol{x}} = A\boldsymbol{x}$ with the matrix above.

Eigenvalues = the natural "speeds" of A:

$$\det(\lambda I - A) = 0$$

1. $\lambda I - A = \begin{bmatrix} \lambda & -1 \\ 4 & \lambda + 2 \end{bmatrix}$
2. $\det = \lambda(\lambda + 2) + 4 = \lambda^2 + 2\lambda + 4$
3. Roots: $\lambda = -1 \pm j1.73$

Eigenvalues of A = poles of the system. This is a mass–spring–damper with $m = 1$, $k = 4$, $c = 2$ ($\omega_n = 2$, $\zeta = 0.5$).

State-space representation

State $\boldsymbol{x}$: the few numbers that describe "where the system is now" (positions, speeds, currents, capacitor voltages).

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + Bu, \qquad y = C\boldsymbol{x} + Du$$

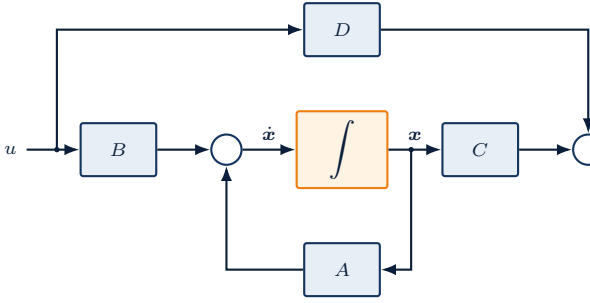
Mass–spring–damper, x_1 = position, x_2 = velocity, $u = F$:

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix}, \; B = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \; D = 0$$

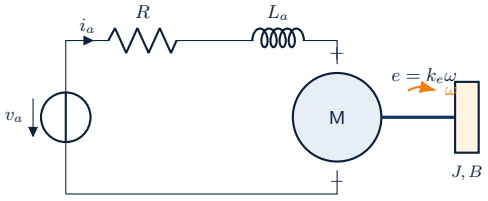
- Handles many inputs/outputs and initial conditions; basis for observers and state feedback (Day 3).

Going further: $G(s) = C(sI - A)^{-1}B + D$; the state choice is not unique; canonical forms, `tf2ss`/`ss2tf`.



One integrator per state variable: integrators "remember" the state.

The DC motor in both representations



$$L_a \frac{di_a}{dt} = v_a - Ri_a - k_e\,\omega \quad \text{(KVL)}$$

$$J \frac{d\omega}{dt} = k_t\, i_a - B\omega \quad \text{(Newton)}$$

SI units: k_t [Nm/A] $= k_e$ [V s/rad].

State space: two energy stores $\Rightarrow$ two states $\boldsymbol{x} = [i_a,\ \omega]^\top$, $u = v_a$, $y = \omega$:

$$A = \begin{bmatrix} -\frac{R}{L_a} & -\frac{k_e}{L_a} \\ \frac{k_t}{J} & -\frac{B}{J} \end{bmatrix},\ B = \begin{bmatrix} \frac{1}{L_a} \\ 0 \end{bmatrix},\ C = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Transfer function (Laplace of both equations, eliminate I_a):

$$\frac{\Omega(s)}{V_a(s)} = \frac{k_t}{(L_a s + R)(Js + B) + k_t k_e}$$

- The $k_t k_e$ term is **back-EMF**: the motor has a **built-in feedback loop** (faster $\Rightarrow$ less current).
- Load torque T_L is a second input (a disturbance).

DC motor: worked numerical example

Small servo motor: $R = 1.2\,\Omega$, $L_a = 2.4\,\text{mH}$, $k_t = k_e = 0.1\,\text{Nm/A}$, $J = 2 \times 10^{-4}\,\text{kg m}^2$, $B = 10^{-5}\,\text{Nm s/rad}$.

1. Denominator divided by $L_a J$:

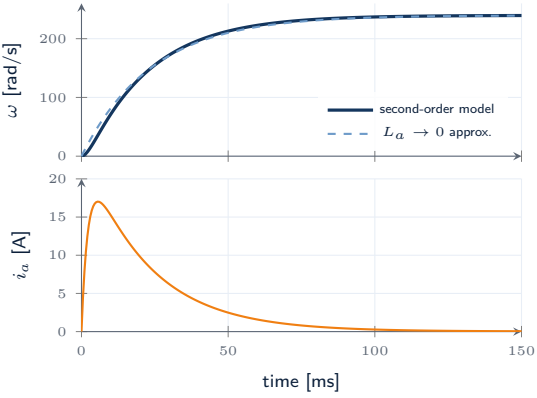
$$s^2 + 500\,s + 20\,860$$

2. Poles: $p_1 = -46\,\text{s}^{-1}$ (mechanical, slow), $p_2 = -454\,\text{s}^{-1}$ (electrical, fast).

3. DC gain: $G(0) = \dfrac{k_t}{RB + k_t k_e} \approx 10\,\text{rad/s}$ per V.

4. 24 V step $\Rightarrow$ final speed $\approx 240\,\text{rad/s}$ (2290 rpm), settles in $\approx 4/46 = 87\,\text{ms}$.

Starting current $\approx 24\,\text{V}/1.2\,\Omega = 20\,\text{A}$!



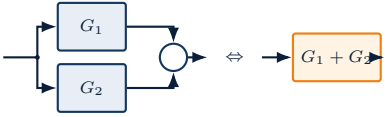
24 V voltage step, no load; computed from the state-space model.

Block-diagram algebra

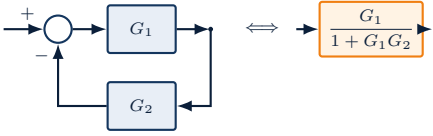
Series



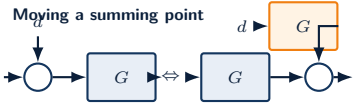
Parallel



Negative feedback



Moving a summing point



Why feedback is powerful

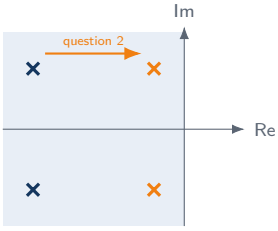
$G_1 = 10$, $G_2 = 1$: $10/11 = 0.91$. Double G_1 to 20: $20/21 = 0.95$.
A 100 % change in G_1 changes the closed loop by only **4 %**.

Positive feedback: $G_1/(1 - G_1G_2)$.
Going further: Mason's gain rule; hidden unstable pole–zero cancellations.

Session 2.2 – Check your understanding

Check your understanding

1. A system has poles at -2 and $-10\,\mathrm{s}^{-1}$. Is it stable? Which pole dominates, and roughly how long does it take to settle?
2. A pair of complex poles moves closer to the imaginary axis (same height). What happens to the step response?
3. Why does the state-space model of the DC motor have exactly *two* states?



Session 2.2 – Summary and key terms

Key take-aways

- Laplace turns ODEs into algebra: $G(s) = Y(s)/U(s)$.
- **Poles** decide stability and speed; left half-plane = stable.
- First order (K, τ) and second order (K, ζ, ω_n) describe most servo sub-systems.
- State space $\dot{\boldsymbol{x}} = A\boldsymbol{x} + Bu$: eigenvalues of A = poles.
- DC motor: back-EMF is built-in feedback; feedback makes gains insensitive to plant changes.

Key terms

Transfer function $G(s)$: output/input in Laplace domain.
Pole / zero: root of denominator / numerator.
Time constant τ: time to reach 63 %.
Damping ratio ζ: how bouncy (0 = no damping).
State: variables that store energy/memory.

Session 3

2.3 System Identification and Frequency Response

Where do models come from? Physics versus measurement

	White box (physics)	**Grey box**	**Black box** (data)
Structure	Newton, Kirchhoff	from physics	generic curve fit
Parameters	data sheets	measured	measured
Example	DC motor model (2.2)	J, B from a run-up test	measured Bode plot of an axis

Identification workflow

1. choose a model shape 2. plan the test (signal, amplitude) 3. record data 4. fit parameters 5. **check** on new data

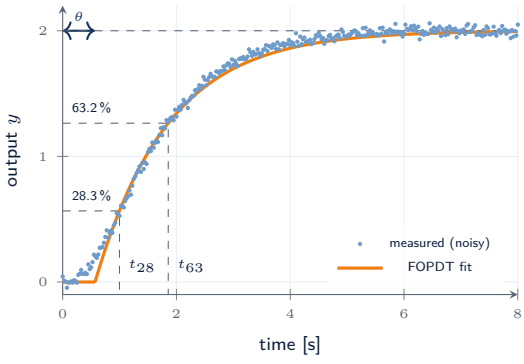
> "A model is never true, only useful" – it must be accurate where it matters for control [Ljung, 1999].



Loxyger, CC BY 4.0, via Wikimedia Commons

HP 3562A signal analyser (1985). Today the analyser is **built into** the servo drive.

Step test: first order plus dead time (FOPDT)



Dead time θ: like a shower – turn the tap and the water temperature changes only seconds later.

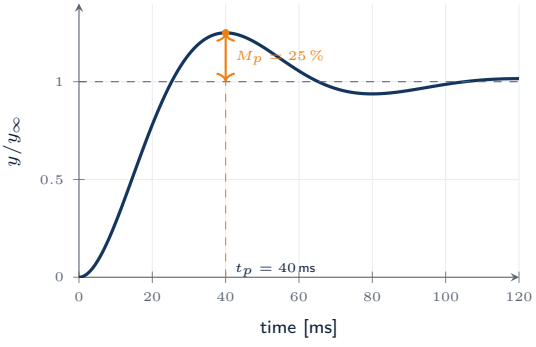
$$G(s) = \frac{K\, e^{-\theta s}}{\tau s + 1}$$

Two-point method, unit step input ($\Delta u = 1$):

1. Final change $\Delta y = 2 \Rightarrow K = \Delta y / \Delta u = 2$.
2. Read $t_{28} = 1.00\,\text{s}$ and $t_{63} = 1.86\,\text{s}$.
3. $\tau = 1.5\,(t_{63} - t_{28}) = 1.5 \cdot 0.86 = 1.29\,\text{s}$.
4. $\theta = t_{63} - \tau = 1.86 - 1.29 = 0.57\,\text{s}$.

Good for thermal/process loops; too coarse for springy mechanics.

Step test: underdamped second-order system



Read the **overshoot** M_p (as a fraction) and the **peak time** t_p:

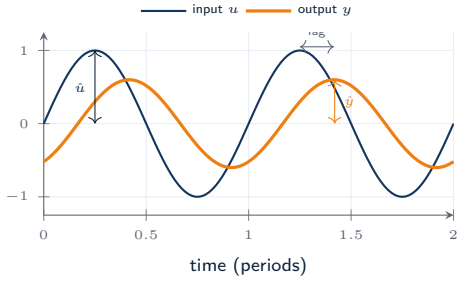
$$\zeta = \frac{-\ln M_p}{\sqrt{\pi^2 + \ln^2 M_p}}, \qquad \omega_n = \frac{\pi/t_p}{\sqrt{1-\zeta^2}}$$

Example (speed loop): $M_p = 0.25$, $t_p = 40\,\text{ms}$

1. $\ln 0.25 = -1.386 \Rightarrow$ $\zeta = 1.386/\sqrt{9.87 + 1.92} = 0.40$
2. $\pi/t_p = \pi/0.04 = 78.5\,\text{rad/s}$
3. $\omega_n = 78.5/\sqrt{1 - 0.40^2} = 86\,\text{rad/s}$ (13.7 Hz)

Going further: logarithmic decrement $\delta = \ln(x_k/x_{k+1})$ for lightly damped ringing.

Refresher: what a sinusoidal frequency response means



Shake a ruler slowly: the tip follows. Shake it fast: the tip moves less and **lags behind**.

Refresher: frequency response

A linear stable system driven by a sine answers with a sine of the **same frequency**, only scaled and shifted.

$$y = |G(j\omega)|\,\hat{u}\,\sin\big(\omega t + \angle G(j\omega)\big)$$

for input $u = \hat{u}\sin\omega t$. **Recipe**: replace s by $j\omega$.

Example $G = \dfrac{1}{s+1}$:

1. $\omega = 1$ rad/s: $G(j1) = \dfrac{1}{1+j} = 0.707\angle -45°$
2. $\omega = 10$ rad/s: $\dfrac{1}{1+10j} = 0.0995\angle -84°$

Faster input $\Rightarrow$ smaller output and more lag.

Refresher: decibels and logarithmic scales

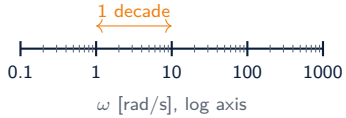
Refresher: the decibel

Gains span from 0.001 to 1000: a **log scale** keeps both visible. Gain in decibels:

$$|G|_{\text{dB}} = 20 \log_{10} |G|$$

- Gains in series multiply $\Rightarrow$ their dB values **add**.
- $-3\,\text{dB} = 0.707$: the usual edge of the "bandwidth".
- A **decade** = frequency $\times 10$.

$\lvert G\rvert$	dB
1000	$+60$
100	$+40$
10	$+20$
2	$+6$
1	0
0.707	-3
0.5	-6
0.1	-20
0.01	-40



Equal steps on the axis = equal *ratios*.

Example: $|G| = 0.05 \Rightarrow 20 \log_{10} 0.05 = -26\,\text{dB}$.

How to read a Bode plot

A **Bode plot** = frequency response for all ω: magnitude in dB (top) and phase in degrees (bottom), both over a log frequency axis.

Reading it:

1. pick ω (e.g. 0.1 rad/s)
2. read magnitude: 40 dB $\Rightarrow$ gain $\times 100$
3. read phase: $\approx -85^\circ$ (output lags)

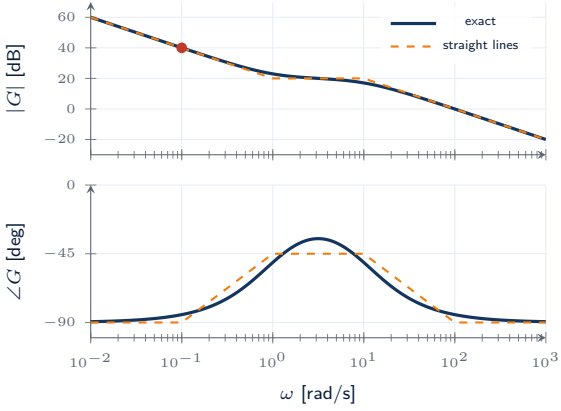
Straight-line rules:

- each pole: -20 dB/decade after its corner, phase drops by 90°
- each zero: $+20$ dB/decade, phase rises by 90°
- integrator $1/s$: -20 dB/decade, -90° everywhere



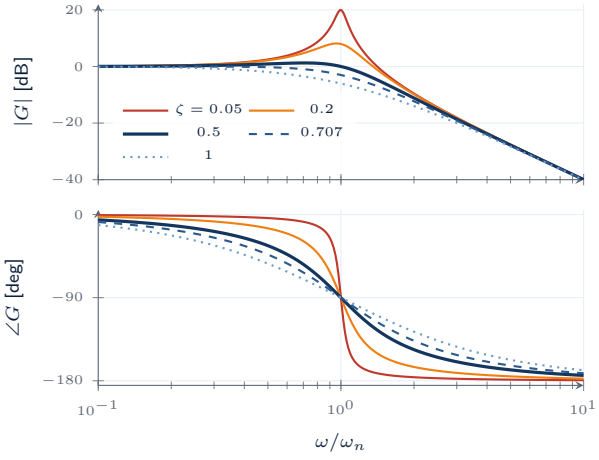
H. W. Bode (1905–1982), Bell Labs. Unknown author, Public domain, via Wikimedia Commons

$$G(s) = \frac{100(s+1)}{s(s+10)}$$



Integrator: -20 dB/dec; zero at 1 flattens; pole at 10 bends down again.

Resonance: pushing a swing at the right moment



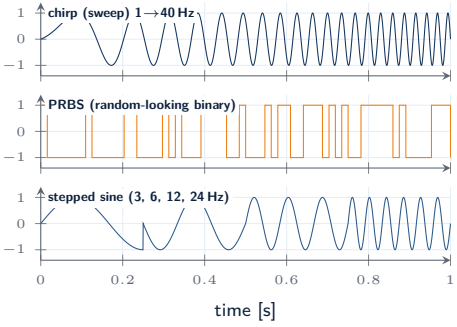
Small pushes at the swing's own rhythm give a **big** motion: that is **resonance**.

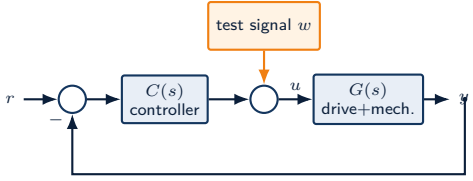
$$|G(j\omega_n)| = \frac{1}{2\zeta}$$

- Below ω_n: output follows (0 dB, 0°).
- At ω_n: peak, phase -90°.
- Above ω_n: falls with -40 dB/decade, phase $\to -180^\circ$.
- $\zeta = 0.05$: peak $1/0.1 = 10$ ($+20$ dB) – typical of a **belt or coupling** resonance.

Going further: peak $M_r = 1/(2\zeta\sqrt{1-\zeta^2})$ at $\omega_n\sqrt{1-2\zeta^2}$; half-power width $\approx 2\zeta\omega_n$.

Measuring a frequency response on a servo axis

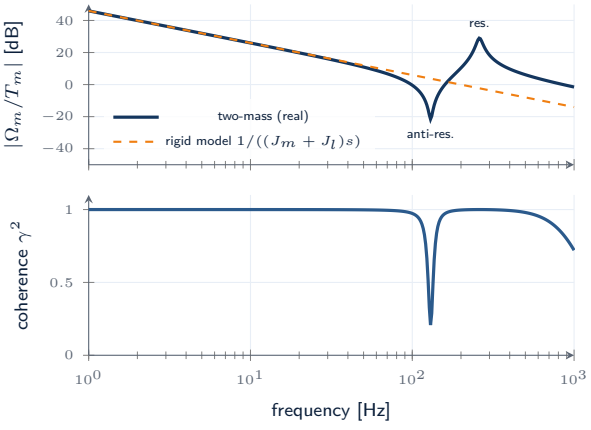




- The drive **adds** a test signal w while the axis is held in closed loop (safe, gravity).
- It records u and y and computes $|G|$ and $\angle G$ at each frequency.
- Amplitude: big enough to overcome friction and noise, small enough to avoid limits.
- Average several runs to reduce noise.

Going further: FRF $\hat{G} = \hat{S}_{wy}/\hat{S}_{wu}$ from cross-spectra; periodic signals avoid leakage.

Measured response of a two-mass axis: coherence and model error



Motor $J_m = 2 \times 10^{-4}$ and load $J_l = 6 \times 10^{-4}\,\mathrm{kg\,m^2}$ joined by a shaft (spring) $k = 400\,\mathrm{Nm/rad}$:

$$f_r = \frac{1}{2\pi}\sqrt{k\,\frac{J_m + J_l}{J_m J_l}} = 260\ \mathrm{Hz}$$

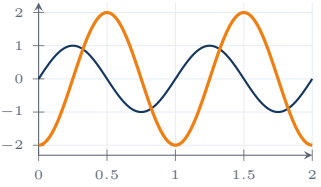
- Up to ≈ 50 Hz the simple rigid model fits; near 130–260 Hz it is **completely wrong**.
- **Coherence** γ^2 $(0 \ldots 1)$: trust the measurement only where $\gamma^2 \gtrsim 0.8$.
- Rule: keep the control bandwidth **below the first resonance** (or add notch filters, Day 3).

Going further: multiplicative uncertainty $G = G_0(1 + \Delta)$; robust stability $|T\,\Delta| < 1$.

Session 2.3 – Check your understanding

Check your understanding

1. Convert a gain of 0.1 and a gain of 1000 to dB.
2. At 10 Hz a 1 V input sine gives a 2 V output sine that lags by a quarter period. What are $|G|$ in dB and $\angle G$?
3. Why must a servo's bandwidth stay below the first mechanical resonance?



Question 2, blue = input, orange = output.

Session 2.3 – Summary and key terms

Key take-aways

- Physics gives the model shape, measurements give the numbers (grey box).
- Step tests: FOPDT (K, τ, θ) or 2nd order $(\zeta, \omega_n$ from M_p, $t_p)$.
- Sine in $\Rightarrow$ sine out: gain $|G(j\omega)|$ and phase $\angle G(j\omega)$.
- Bode plot: dB and degrees over log frequency; $-20\,\text{dB/dec}$ per pole.
- Resonances make simple models wrong $\Rightarrow$ limit the bandwidth.

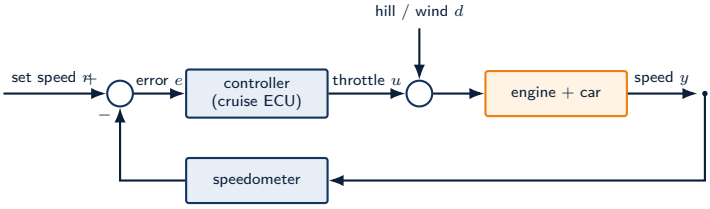
Key terms

Dead time θ: pure delay before anything happens.
Frequency response: gain and phase vs. frequency.
Decibel: $20 \log_{10}$ of a gain.
Resonance: large response near ω_n.
Coherence: measurement quality, $0 \ldots 1$.

Session 4

2.4 PID and Basic Control Structures

Feedback: the cruise-control story



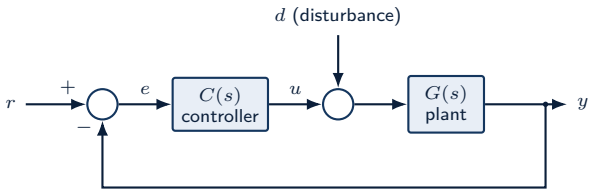
- **Open loop** (fixed throttle): a hill slows the car and nobody notices.
- **Closed loop**: **measure** speed, **compare** with set speed, **correct** the throttle.
- Feedback fights disturbances and model errors *it does not know about*.

P control only

Controller $u = K_p\, e$. Car: 2 km/h per % throttle ($G = 2$); $K_p = 4.5\,\%/(\text{km/h}) \Rightarrow L = K_p G = 9$.

1. $y = \frac{L}{1+L}\, r = \frac{9}{10} \cdot 100 = 90\,\text{km/h}$
2. A **10 km/h offset** remains – P alone needs an error to push.

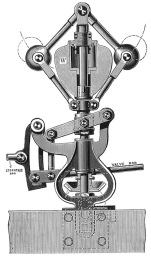
The standard feedback loop and its two key functions



Loop gain $L(s) = C(s)G(s)$: go once around the loop. **Sensitivity** S: what is left of a disturbance; **complementary sensitivity** T: how well y follows r.

$$S = \frac{1}{1+L}, \qquad T = \frac{L}{1+L}$$

- Cruise control: $L = 9 \Rightarrow S = 0.1,\ T = 0.9;\ S + T = 1$ always.
- Big $|L| \Rightarrow$ small $S \Rightarrow$ good tracking – **but** sensor noise also reaches y.

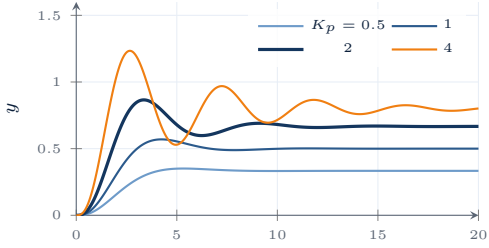


Heat Engines (1913), scan by Andy Dingley, Public domain, via Wikimedia Commons

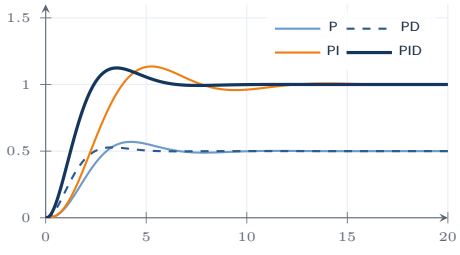
Watt's centrifugal governor: a mechanical P controller for steam-engine speed. Maxwell's paper "On Governors" (1868) started control theory.

P, I and D: present, past and future of the error

P only, $G = 1/(s+1)^3$



same plant, four controllers



$$u(t) = K_p\, e + K_i \int_0^t e\, d\tau + K_d\, \dot{e}$$

P = now, **I** = accumulated past, **D** = trend (future).

More ...	speed	overshoot	steady-state error
P (K_p)	faster	larger	smaller, not zero
I (K_i)	faster	larger	**removed**
D (K_d)	similar	smaller	no effect (noise ↑)

PID forms: same idea, different parameters

Form	Transfer function	Typical use
Parallel	$C = K_p + \dfrac{K_i}{s} + K_d s$	textbooks, drives
Ideal (ISA)	$C = K_p\left(1 + \dfrac{1}{T_i s} + T_d s\right)$	process controllers
Series	$C = K'_p\left(1 + \dfrac{1}{T'_i s}\right)\left(1 + T'_d s\right)$	older PLCs

Conversion parallel $\leftrightarrow$ ideal: $K_i = K_p/T_i$, $K_d = K_p T_d$.
T_i = integral time [s], T_d = derivative time [s].

- Example: $K_p = 2$, $T_i = 4\,\text{s}$, $T_d = 0.5\,\text{s} \Rightarrow K_i = 0.5\,\text{s}^{-1}$, $K_d = 1\,\text{s}$.
- **Pitfall**: copying gains between devices without checking form and units (s vs. min, % vs. A).

Going further: series $\rightarrow$ ideal conversion formulas.



Bitjungle, CC BY-SA 4.0, via Wikimedia Commons

Industrial PID controllers usually ask for K_p, T_i, T_d (ideal form).

Classical tuning: Ziegler–Nichols (1942)

Recipe: use P only, raise K_p until the loop just keeps oscillating. Note this **ultimate gain** K_u and the period T_u.

	K_p	T_i	T_d
P	$0.5K_u$	–	–
PI	$0.45K_u$	$T_u/1.2$	–
PID	$0.6K_u$	$T_u/2$	$T_u/8$

Going further: step-response version: PID $K_p = 1.2\tau/(K\theta)$, $T_i = 2\theta$, $T_d = \theta/2$.

Worked example

Plant $G = 1/(s+1)^3$.

1. Each pole lags up to $90°$; total $-180°$ when $3\arctan\omega = 180° \Rightarrow \omega_u = \sqrt{3} = 1.73$ rad/s.
2. There $|G| = 1/(1+3)^{3/2} = 1/8 \Rightarrow K_u = 8$.
3. $T_u = 2\pi/1.73 = 3.63$ s.
4. PID: $K_p = 4.8$, $T_i = 1.81$ s, $T_d = 0.45$ s.

Caution

Gives fast but **aggressive** loops (big overshoot). Pushing a real machine to oscillation can be unsafe. Use ZN only as a starting point [Åström & Hägglund, 2006].

Model-based tuning: one knob for speed

For an FOPDT model $Ke^{-\theta s}/(\tau s+1)$ and a PI controller, choose the **desired closed-loop time constant** τ_c:

$$K_p = \frac{\tau}{K(\tau_c + \theta)}, \qquad T_i = \tau$$

- Smaller $\tau_c \Rightarrow$ faster but less robust.
- SIMC default $\tau_c = \theta$: good trade-off [Skogestad, 2003].
- Lambda tuning: same formula, $\tau_c = \lambda$, often $\lambda = \tau$ (gentle).

Going further: SIMC $T_i = \min\{\tau, 4(\tau_c + \theta)\}$; Cohen–Coon gives $K_p = 1.06$, $T_i = 1.0\,\text{s}$ here (aggressive).

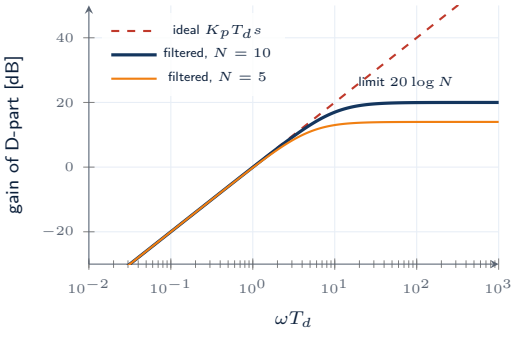
Example: FOPDT from session 2.3

$K = 2$, $\tau = 1.29\,\text{s}$, $\theta = 0.57\,\text{s}$

1. SIMC, $\tau_c = \theta = 0.57\,\text{s}$:
   $$K_p = \frac{1.29}{2\,(0.57 + 0.57)} = 0.57$$
2. Lambda, $\lambda = \tau = 1.29\,\text{s}$:
   $$K_p = \frac{1.29}{2\,(1.29 + 0.57)} = 0.35$$
3. Both: $T_i = \tau = 1.29\,\text{s}$
4. ZN step rule (PI): $K_p = 0.9\tau/(K\theta) = 1.02$

A 3:1 spread in K_p: the rules aim at **different robustness**, there is no single "right" answer.

Derivative action: always filter it

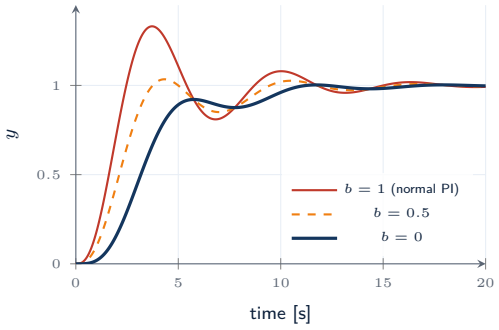


A pure derivative amplifies fast wiggles without limit: sensor noise becomes huge.

$$D(s) = \frac{K_p T_d\, s}{1 + T_d s/N}, \qquad N \approx 5 \ldots 20$$

- The filter caps the high-frequency D-gain at $K_p N$.
- **Derivative on measurement** ($-\dot{y}$ instead of $\dot{e}$): no "kick" when the set-point jumps.
- Servo practice: D on position = velocity feedback; in a cascade it is the inner speed loop.
- Example: 20-bit encoder, 125 μs sampling: one count gives a speed jump of 0.048 rad/s.

Set-point weighting: gentle on commands, firm on disturbances



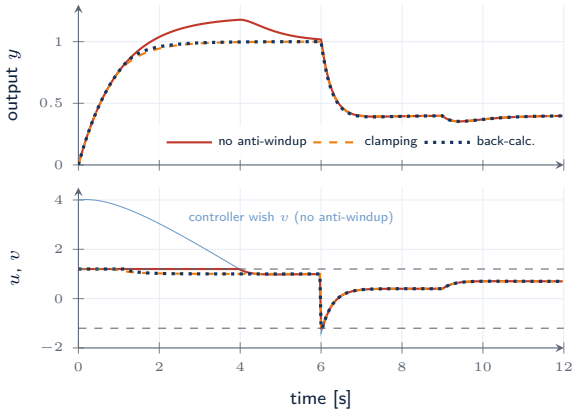
PI, $K_p = 2$, $T_i = 2.5\,\text{s}$, $G = 1/(s+1)^3$. Overshoot: 33 %, 3.4 %, 0.3 %.

Use only part (b) of the set-point in the P-term:

$$u = K_p\left[(b\,r - y) + \frac{1}{T_i}\int (r - y)\,dt\right]$$

- The I-term keeps the **true** error, so the final value is still exact.
- Disturbance response and stability are **unchanged**: b only shapes the reaction to new set-points.
- Recipe: tune K_p, T_i for disturbances first, then pick b.
- Servo equivalent: smooth trajectory + feedforward (next slides).

Saturation and integrator windup – and the cure



Plant $1/(s+1)$, PI $K_p = 4$, $T_i = 0.8\,\text{s}$, limit $|u| \le 1.2$.

Cruise control uphill: throttle is at 100 % but the car is still too slow.

- The integrator keeps adding up error although the throttle **cannot** open further: it "winds up".
- On the hill-top it must first "unwind" ⇒ big overshoot (18 % here) and slow settling.

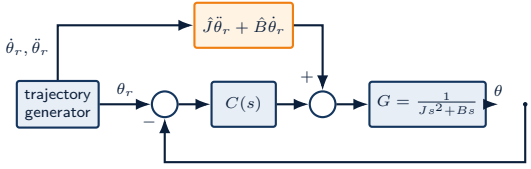
Anti-windup

Clamping: stop integrating while the output is at its limit and the error pushes further. Simple, standard in drives.

Back-calculation: feed $(u - v)/T_t$ back into the integrator so it tracks the limit ($T_t \approx T_i/2$).

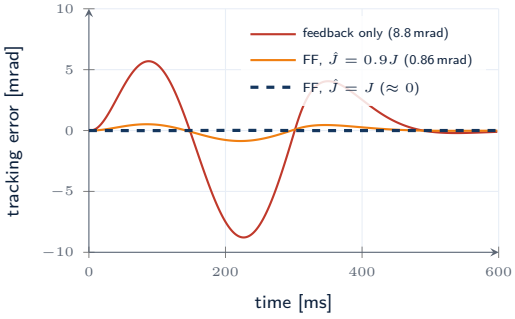
Going further: in cascades, pass the inner loop's limit to the outer integrator.

Feedforward: let feedback correct only what is unknown



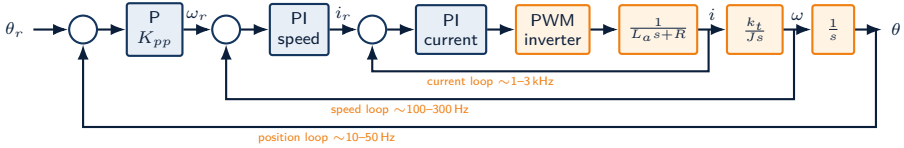
A driver who *sees* the hill presses the pedal **before** the car slows: that is feedforward.

$$T_{\mathrm{FF}} = \hat{J}\,\ddot{\theta}_r + \hat{B}\,\dot{\theta}_r$$



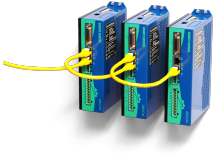
- Even with a 10 % inertia error: $10\times$ smaller error.
- FF sits outside the loop: it does **not** affect stability.
- Needs a smooth reference (S-curve profile).

Cascade control: the servo drive structure



Loop	Simple rule	DC motor of session 2.2
Current PI	$K_p = L_a\omega_{cc}$, $T_i = L_a/R$	1 kHz: $K_p = 15.1\,\mathrm{V/A}$, $T_i = 2\,\mathrm{ms}$
Speed PI	$K_p = J\omega_{sc}/k_t$, $T_i \approx 4/\omega_{sc}$	100 Hz: $K_p = 1.26\,\mathrm{A\,s/rad}$, $T_i = 6.4\,\mathrm{ms}$
Position P	$K_{pp} \approx \omega_{sc}/4\ldots/10$	$K_{pp} = 126\,\mathrm{s}^{-1}$ (20 Hz)

- Like a company: the fast inner loop does the job the outer loop asks for. Tune **inside-out**.
- Each outer loop 4–$10\times$ slower than the one inside it. Details on Day 4.



Tero Kontkanen, CC BY-SA 3.0, via Wikimedia Commons

Session 2.4 – Check your understanding

Check your understanding

1. Cruise control with P only, driving up a long hill: why does the speed stay a little too low? Which term fixes it?
2. Why do we always filter the D-term?
3. The throttle is fully open for 30 s on a steep hill. What happens inside the integrator, and what do you see at the top of the hill?



disturbance d = hill

Session 2.4 – Summary and key terms

Key take-aways

- Feedback = measure, compare, correct; it fights unknown disturbances.
- P reacts to now, I removes the steady error, D damps (filter it!).
- Tune with a model (τ_c knob) rather than by pushing to instability.
- Every saturating integrator needs **anti-windup**.
- Feedforward and set-point weighting improve tracking without changing stability.
- Servo drives: current $\rightarrow$ speed $\rightarrow$ position cascade, tuned inside-out.

Key terms

Loop gain $L = CG$: gain once around the loop.
Sensitivity $S = 1/(1 + L)$: fraction of disturbance left.
Windup: integrator grows while the actuator is at its limit.
Feedforward: act on the known command before an error appears.
Cascade: loops nested inside each other.

Session 5

2.5 Stability and Performance Analysis

What does “stable” mean?



stable
returns

unstable
runs away

marginal
stays where pushed

Shower analogy: the hot water arrives late (dead time). If you over-react to what you feel now, the temperature swings ever more – **feedback can create instability**.

Closed-loop poles are the roots of the **characteristic equation** $1 + L(s) = 0$.

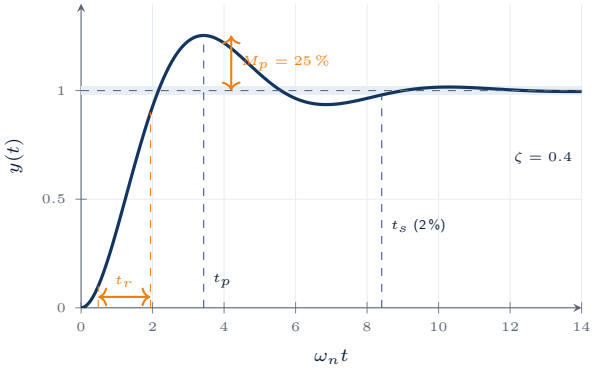
stable $\iff$ all closed-loop poles have $\mathrm{Re}(p) < 0$

Example $L(s) = \dfrac{K}{s(s+2)(s+5)}$:

1. $1 + L = 0 \Rightarrow s^3 + 7s^2 + 10s + K = 0$
2. $K = 20$: poles -5.88 and $-0.56 \pm j1.76 \Rightarrow$ stable
3. $K = 100$: two poles in the right half-plane $\Rightarrow$ unstable

Too much gain destabilises – we will see why on the next slides.

Time-domain specifications



Measure	Meaning
Rise time t_r	$10\,\% \to 90\,\%$
Overshoot M_p	how far past the target
Settling time t_s	stays within $\pm 2\,\%$
Steady-state error	error left at the end

For a 2nd-order closed loop, ζ **alone fixes the overshoot**:

$$M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}, \qquad t_s \approx \frac{4}{\zeta\omega_n}$$

ζ	0.3	0.4	0.5	0.6	0.7
M_p [%]	37	25	16	9.5	4.6
PM [deg]	33	43	52	59	65

Rule of thumb: phase margin $\approx 100\,\zeta$ [deg].

Steady-state error and system type

Final value theorem (only for a **stable** loop):

$$e_\infty = \lim_{t\to\infty} e(t) = \lim_{s\to 0} s\,E(s), \qquad E = \frac{R}{1+L}$$

System type = number of integrators $(1/s)$ in $L(s)$.

Type	step	ramp	example
0	$\frac{1}{1+L(0)}$	∞	cruise control, P only
1	0	$\frac{1}{K_v}$	PI, or position loop
2	0	0	position loop + I

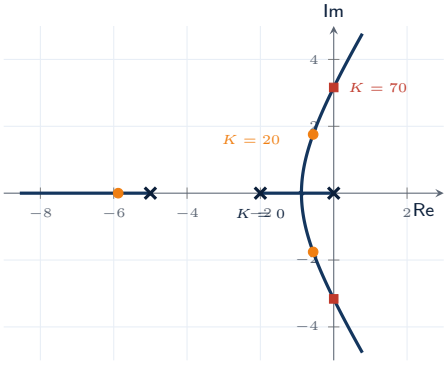
$K_v = \lim_{s\to 0} sL(s)$ [s^{-1}], the velocity constant.

Servo example: following error

1. Position P gain $K_{pp} = 126\,\mathrm{s}^{-1}$ around a fast speed loop: $L \approx K_{pp}/s \Rightarrow$ type 1, $K_v = 126\,\mathrm{s}^{-1}$.
2. Table moves at 0.2 m/s, 10 mm ball-screw pitch $\Rightarrow$ motor speed $\dot\theta = 126$ rad/s (a ramp in θ).
3. $e_\infty = \dot\theta/K_v = 126/126 = 1$ rad.
4. 1 rad $\times 10\,\mathrm{mm}/2\pi = 1.6$ mm lag behind the command.

Cure: velocity feedforward (previous session) removes this lag.

Root locus: how closed-loop poles move with gain



Same loop $L = \dfrac{K}{s(s+2)(s+5)}$; the curves show where the 3 closed-loop poles go as K grows from 0.

- $K = 0$: poles start at the open-loop poles $0, -2, -5$ ($\times$).
- Small K: two poles meet and split into a complex pair $\Rightarrow$ oscillation starts.
- $K = 20$: $-0.56 \pm j1.76$ ($\zeta = 0.30$, $\approx 37\,\%$ overshoot).
- $K = 70$: poles reach the $j\omega$-axis at $\pm j3.16$ $\Rightarrow$ **stability limit**.
- $K > 70$: unstable.

Going further: sketching rules: asymptote centroid -2.33, angles $\pm 60^\circ$, break-away at -0.88.

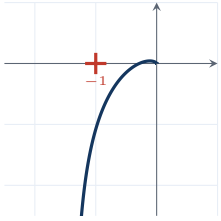
Going further: Routh–Hurwitz and Nyquist (concepts only)

Routh–Hurwitz

Decides stability from the polynomial coefficients without solving for the roots. For $s^3 + 7s^2 + 10s + K$:

s^3	1	10
s^2	7	K
s^1	$(70 - K)/7$	0
s^0	K	

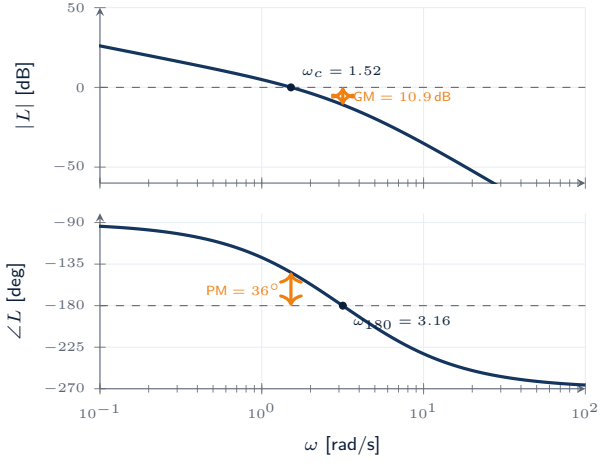
No sign change in the first column $\iff$ $0 < K < 70$: the same limit as the root locus.



Nyquist: plot $L(j\omega)$ in the complex plane. For a stable open loop, the closed loop is stable if the curve does **not encircle** -1.

The distance of the curve from -1 is a measure of robustness. Nyquist also works with dead time and with measured data. The margins on the next slide are a simple reading of this idea.

Gain and phase margins: how far from the edge?

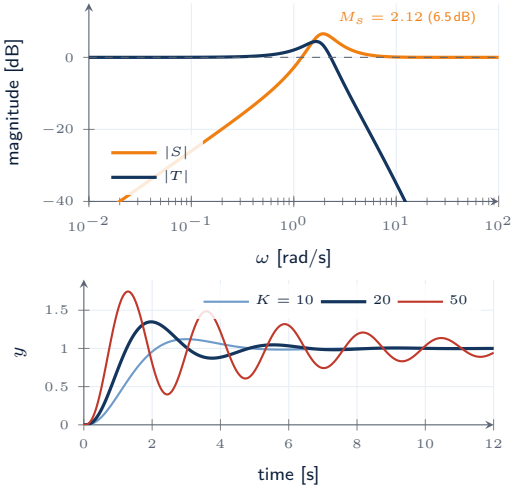


The loop oscillates if a signal comes back around the loop with gain 1 and $-180°$. Margins = **safety distance** from that point.

1. Find ω_c where $|L| = 0\,\text{dB}$ ($1.52\,\text{rad/s}$).
2. Read the phase there: $-144° \Rightarrow$ **phase margin** PM $= 180° - 144° = 36°$.
3. Find where the phase is $-180°$ ($3.16\,\text{rad/s}$); $|L| = -10.9\,\text{dB}$ there $\Rightarrow$ **gain margin** GM $= 10.9\,\text{dB}$ ($\times 3.5$).

- Targets: PM 45–$60°$, GM $\geq 6\,\text{dB}$.
- A delay T eats phase $\omega_c T$: here $0.63\,\text{rad}/1.52 = 0.41\,\text{s}$ delay would make it oscillate.

Sensitivity, robustness and loop shaping



$|S(j\omega)|$ = how much of a disturbance at frequency ω survives the loop.

$$M_s = \max_{\omega} |S(j\omega)| \quad \text{(aim: } 1.2 \ldots 2\text{)}$$

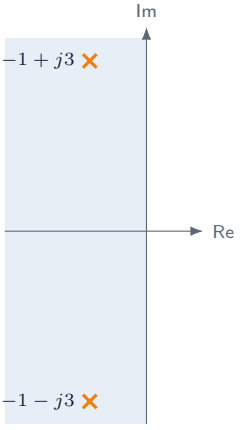
- Low frequency: $|S| \ll 1 \Rightarrow$ disturbances removed.
- Near crossover: $|S| > 1 \Rightarrow$ feedback **amplifies** disturbances. Peak $M_s = 1/$(closest distance to -1).
- $K = 20$: $M_s = 2.1$, $36°$ PM $\Rightarrow$ ringing; $K = 10$ is calmer.
- **Loop shaping**: make $|L|$ big at low frequency (tracking), small at high frequency (noise, resonances).

Going further: $\mathrm{GM} \geq M_s/(M_s - 1)$, $\mathrm{PM} \geq 2\arcsin(1/2M_s)$; waterbed effect $\int_0^\infty \ln|S|\, d\omega = 0$.

Session 2.5 – Check your understanding

Check your understanding

1. Closed-loop poles at $-1 \pm j3$. Stable? Small or large overshoot?
2. A loop has a phase margin of only 10°. What would you see in its step response?
3. A type-1 position loop has $K_v = 100\,\mathrm{s}^{-1}$ and moves at 50 rad/s. What is the following error?



Session 2.5 – Summary, design targets and key terms

Key take-aways

- Stable $\Longleftrightarrow$ closed-loop poles in the left half-plane.
- ζ sets overshoot; PM $\approx 100\zeta$.
- System type sets the steady-state error.
- More gain $\Rightarrow$ poles move right, margins shrink.

Typical design targets

Quantity	Target
Phase margin	45–60°
Gain margin	≥ 6 dB
M_s	1.2–2.0
Overshoot (position)	0–5 %
ω_c vs. resonance	$< \omega_r/3$

Our $K = 20$: PM $36°$, $M_s = 2.1$ $\Rightarrow$ too aggressive; $K \approx 10$ meets the targets.

Key terms

Characteristic eq. $1 + L = 0$.
Crossover ω_c: $|L| = 1$.
Phase margin: extra lag tolerated.
Gain margin: extra gain tolerated.
System type: integrators in L.

Session 6

Wrap-up

Day summary and preview of Day 3

Modelling chain

nonlinear physics $\xrightarrow{\text{tangent}}$ linear ODE $\xrightarrow{\text{Laplace}}$ $G(s)$ or $(A, B, C, D) \xrightarrow{s=j\omega}$ Bode plot

. . . or measure it: step test / frequency response.

Control chain

feedback (cruise control) → PID + filter + anti-windup + feedforward → cascade current/speed/position → check poles, PM, GM, M_s.

Numbers to remember

63 % at $t = \tau$ ▪ $\zeta = 0.7$: 5 % overshoot ▪ -20 dB/dec per pole ▪ PM 45–60° ▪ GM ≥ 6 dB

Day 3 – filters, modern, adaptive and discrete control

- Filters in the loop: low-pass and notch, and their phase cost.
- State feedback and observers (built on today's A, B, C, D).
- Discrete-time control: sampling and computation delay.
- Servo architecture: inverter and DC link.



Pieceofmetalwork, CC BY-SA 4.0, via Wikimedia Commons

Strain-wave gear: springy transmissions motivate notch filters.

Figure credits

All other drawings and plots: own work (TikZ / pgfplots; numerical data from `day2/data/gen_data.py`).

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